# ED Waiting Time Dataset Dictionary

Healthcare Operations & Patient Risk Analytics Capstone | Reference guide for students

## Dataset overview

The supplied file contains **6,000 rows** and **19 columns**. Each row represents a patient visit. This dictionary explains the fields, their intended analytical use, and important data-quality cautions. Descriptions follow the capstone project brief and the supplied file; where the source does not establish the meaning of a value, students should investigate rather than assume.

## Field-by-field dictionary

| Column name | Category | Meaning | Format / expected values | Quality and analysis notes |
|---|---|---|---|---|
| **patient_id** | Identifier | Unique identifier for a patient visit. | Text/string; e.g., PT00001 | Unique per visit; exclude from predictive features. |
| **visit_date** | Date/time | Calendar date on which the visit occurred. | Date; YYYY-MM-DD | Parse as a date before time-based analysis. |
| **day_of_week** | Calendar | Name of the weekday on which the visit occurred. | Category; Monday–Sunday | Can be cross-checked against visit_date. |
| **day_type** | Calendar | Groups visits into Weekday or Weekend. | Category; Weekday / Weekend | Check consistency with day_of_week. |
| **season** | Calendar/context | Seasonal period assigned to the visit date. | Category; e.g., Harmattan/Dry, Peak Rainy | Use categories as provided; confirm definitions with dataset documentation. |
| **arrival_time** | Time | Time of day when the patient arrived. | Time; HH:MM | Use with visit_date to derive a datetime if needed. |
| **arrival_hour** | Time-derived | Hour of arrival extracted from arrival_time. | Integer; 0–23 expected | Zero means midnight hour (00:00–00:59), not missing by default. |
| **department** | Operational | Department where the patient was seen or routed. | Category; Emergency, Outpatient - General, Pediatrics, Maternity, Surgery Referral | Compare waiting times and patient volumes across departments. |
| **triage_priority** | Clinical/operational | Urgency category assigned during triage; lower numbers indicate higher urgency. | Integer; 1–5 in the project brief | Priority 1 is most urgent; check whether high-acuity patients experience delays. |
| **chief_complaint** | Clinical text | Main symptom or reason for the visit, as recorded. | Text/category | Has approximately 2% missing values by design; investigate before imputation. |
| **age** | Demographic | Patient age in years. | Integer; observed values should be checked against context | Zero may represent an infant under one year or a coding issue. Investigate; do not automatically replace. |
| **gender** | Demographic | Gender category recorded for the visit. | Category; values observed in the file | Check distinct values and missingness; preserve source categories. |
| **staff_on_duty** | Operational | Number of clinical staff on duty at the time of arrival. | Count; integer | Zero would need validation against staffing records and definitions. |
| **patients_in_queue_at_arrival** | Operational | Number of patients already waiting when the patient arrived. | Count; integer; zero can be valid | Zero may mean no queue. Check definition and records before treating zero as missing. |
| **registration_time_minutes** | Process duration | Time spent completing registration. | Numeric; minutes | Check non-negative values and unusual outliers. |
| **triage_time_minutes** | Process duration | Time spent in the triage process. | Numeric; minutes | Check non-negative values and unusual outliers. |

| Column name | Category | Meaning | Format / expected values | Quality and analysis notes |
|---|---|---|---|---|
| **wait_time_to_provider_minutes** | Outcome/target | Time from triage to being seen by a provider; the primary waiting-time outcome. | Numeric; minutes | Zero may be a valid immediate wait or a data issue. Confirm the source definition before changing it. |
| **consultation_duration_minutes** | Process duration | Time spent with the provider during consultation. | Numeric; minutes | Occurs after provider contact; exclude from models predicting the preceding wait time. |
| **total_visit_duration_minutes** | Process duration | Total end-to-end duration of the visit. | Numeric; minutes | May overlap with or include the target and later processes; usually exclude from models predicting wait_time_to_provider_minutes to avoid leakage. |

## Data quality and modeling guidance

Do not treat every zero as missing.

arrival_hour = 0 is a valid hour (midnight). A zero patients_in_queue_at_arrival may mean nobody was waiting. Zero age may represent an infant under one year or a coding issue. A zero provider wait may be genuine or may reflect a recording convention. Verify each meaning from the dataset definition and related records.

NaN conversion and imputation are different steps.

First establish that a value is missing/invalid. Only then decide whether to leave it missing, exclude it for a specific calculation, or impute it. Do not replace a minimum with the mean simply because it is the minimum.

Check missingness and duplicates.

Inspect missing-value counts, duplicate rows, data types, distinct categories, valid ranges, and possible inconsistencies between visit_date, day_of_week, day_type, arrival_time, and arrival_hour. The chief_complaint field is expected to have about 2% missing values in this educational dataset.

Avoid target leakage.

For a model predicting wait_time_to_provider_minutes, do not use consultation_duration_minutes or total_visit_duration_minutes as predictors because they occur after or overlap the waiting-time outcome. Do not use patient_id as a predictive feature.

Document every decision.

Keep the raw file unchanged. Create a cleaned copy, record each change and its justification, and compare key statistics before and after cleaning. Distinguish confirmed data errors from valid but unusual observations.

Educational-use note.

This dataset is synthetic and intended for learning and portfolio projects. It is not a source of clinical evidence or a basis for real patient-care decisions.

## Suggested first-pass Python checks

```
df.shape
df.info()
df.isna().sum()
df.duplicated().sum()
df.describe(include='all')
df['arrival_hour'].value_counts(dropna=False).sort_index()
df['patients_in_queue_at_arrival'].value_counts(dropna=False).sort_index()
```